

SVKM'S NMIMS

**MUKESH PATEL SCHOOL OF TECHNOLOGY MANAGEMENT & ENGINEERING/
SCHOOL OF TECHNOLOGY MANAGEMENT**

Academic Year: 2024-2025

Program/s: BTech/MBATech/BTI

Year: II/IV Semester: IV/VIII

Stream/s ;Data Science/CSE DS(7057)

Subject: Introduction to Data, Signal and Image Analysis

Time: _03 hrs (10 am to 1pm)

Date: 10 / 05 / 2025

No. of Pages: _03_

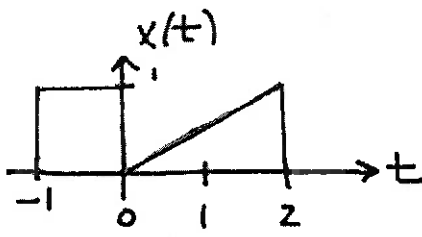
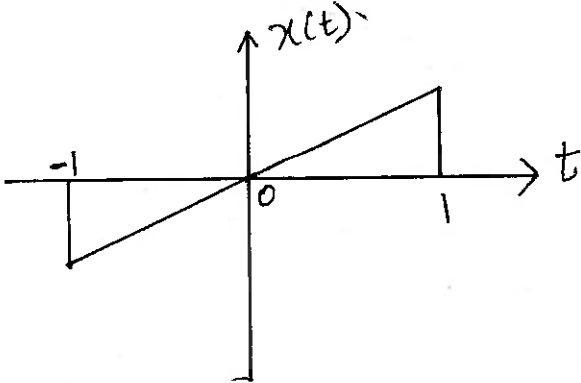
Marks: __100__

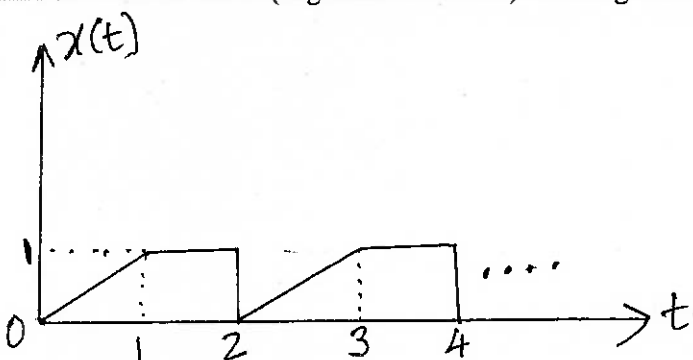
Final Examination 2024-25/Re-exam 2023-24

Instructions: Candidates should read carefully the instructions printed on the question paper and on the cover of the Answer Book, which is provided for their use.

- 1) Question No. _1_ is compulsory.
- 2) Out of remaining questions, attempt any _4_ questions.
- 3) In all _5_ questions to be attempted.
- 4) All questions carry equal marks.
- 5) Answer to each new question to be started on a fresh page.
- 6) Figures in brackets on the right-hand side indicate full marks.
- 7) Assume Suitable data if necessary.

Q1		Answer briefly:	[20]
CO-1, BL-1; CO-1, BL-5, SO1	a.	Find Z- Transform of $x(n) = 2, 3, 1$	[5]
CO-1, BL - 3, SO2	b.	Solve to find initial value and final value for the given transfer function. $X(S) = \frac{4S+2}{s(S+4)}$	[5]
Co-2 , BL-1; SO1	c.	List the importance of FIR filter	[5]
Co1 BL2, SO1	d.	Define Pixel and Pixel Density	[5]
Q2 CO-1, BL-1; CO-3, BL-3, SO1	A	Explain the significance of sampling to convert continuous time into discrete. Apply sampling concept on a continuous time signal $x(t) = 5 \cos(10\pi t)$ and find it's discrete signal	[10]

Q2 Co1,BL3, SO1	B	A continuous-time signal is shown below. Sketch the following signal:- $x(-t)$, $x(2t)$, $x(t-2)$, $3x(t)$ 	[10]
Q3 CO-3, BL-5, SO2	A	Determine $Y(t)$ by using integral convolution. Given - $x(t) = 3e^{-2t}u(t) \text{ and } h(t) = e^{-t}u(t)$	[10]
Q3 CO-1, BL-5 SO2	B	Evaluate- $L[\int_0^t t \cdot e^{3t} \cos 4t \, dt]$	[10]
Q4 Co1 BL1 SO1	A	Plot and write equations for basic (standard) continuous unit $U(t)$ and ramp $r(t)$ signals. Hence plot: $-5U(t+5)$ and $r(t+3)$.	[10]
Q4 CO-1, BL-1 SO1	B	Evaluate Laplace Transform for the following signal $x(t)$: 	[10]
Q5 CO-2, BL-2 SO1	A	Discuss the advantages, disadvantages and applications of IIR filter.	[10]

Q5 Co4 BL1,SO1	B	Write a detailed note on image compression.	[10]
Q6 CO-2, BL-4 SO1	A	Differentiate FIR filter vs IIR filter	[10]
Q6 Co4 BL5	B	"Image enhancement is performed to improve the visual quality of an image by adjusting its brightness, contrast, sharpness, or reducing noise". Justify the statement with detailed explanation.	[10]
Q7 Co4, BL1,BL2	A	Give a Summary on JPG, PNG, GIF, and TIFF.	[10]
Q7 CO-1, BL-3	B	Solve to find the Fourier series (trigonometric form) for the given signal 	[10]

